LINHAO BAI

Robotics Software | Robot Learning | Sim-to-Real | Research Engineering

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GitHub | Scholar | LinkedIn

EXPERIENCE

Tesla - Robotics Integrations Engineer Sep 2026 - Present

Austin, TX

- Integrate production AMR and ASRS software across fleet orchestration, task sequencing, localization, robot runtime / firmware, and PLC / OPC UA interfaces to factory equipment.
- Develop and debug multi-robot resource constraints, recovery workflows, and station handoffs; validate physical execution with robot telemetry, Python tooling, and distributed-service logs.

Georgia Institute of Technology - STAR Lab Oct 2024 - May 2026

Graduate Research Assistant, Robot Learning | Atlanta, GA

- Built a ROS learned-policy-to-hardware interface for a Kinova arm and PSYONIC / Allegro dexterous hands, integrating SDKs, joint feedback, nonblocking velocity commands, synchronization, timeouts, and logging.
- Modified and evaluated TD-MPC / TD-MPC2 world-model dynamics; compared MLP, linear, and SINDy-regularized variants using prediction error and planning returns to investigate dynamics-planner mismatch.
- Extended Diffusion Policy training and evaluation for DexArt: object-flow datasets, custom training workspace, checkpoint loading, and offline action-error evaluation.
- Used MuJoCo, NVIDIA Isaac Sim / Isaac Gym for robot-learning baselines and sim-to-real experiments; contributed to Video2Sim2Real perception, residual RL, and real-robot execution.

SEER Robotics - Software Engineer / Project Manager Jun 2022 - Jul 2024

Shanghai, China

- Developed C++ robot-controller services and integration plugins for OTA / backup, time synchronization, health monitoring, and telemetry; debugged firmware, LiDAR, localization, and robot interfaces.
- Delivered AMR solutions to 35+ international clients; led a 5-AMR / 100+ rack warehouse system with SLAM + QR correction, reservations, and randomized stress tests across hundreds of moves.

SELECTED ROBOTICS RESEARCH

K-UBM | CoRL 2026 | Equal contribution [paper]
Going with the Flow: visual-flow / Koopman latent dynamics for long-horizon dexterous behavior and event-triggered replanning.

Video2Sim2Real | 2026 preprint [paper]
Video-to-simulation-to-real dexterous skill acquisition, spanning scene reconstruction, residual RL, and learned-policy execution on physical hardware.

EDUCATION

Georgia Institute of Technology - M.S. Computer Science, GPA 4.0 / 4.0 May 2026

Chongqing Jiaotong University - B.Eng. Civil Engineering Jun 2022

TECHNICAL SKILLS

Programming / Systems: Python, C++, Bash, Linux, Git, Docker, Kubernetes, logging / telemetry, debugging
Robotics / Hardware: ROS / ROS2, Kinova & hand SDKs, actuator control, sensor feedback, CAN, RS-485, OPC UA, LiDAR, SLAM, calibration
Simulation / ML: MuJoCo, NVIDIA Isaac Sim, Isaac Gym, PyTorch, TD-MPC, MPPI, SINDy, Koopman, Diffusion Policy, imitation / reinforcement learning, sim-to-real